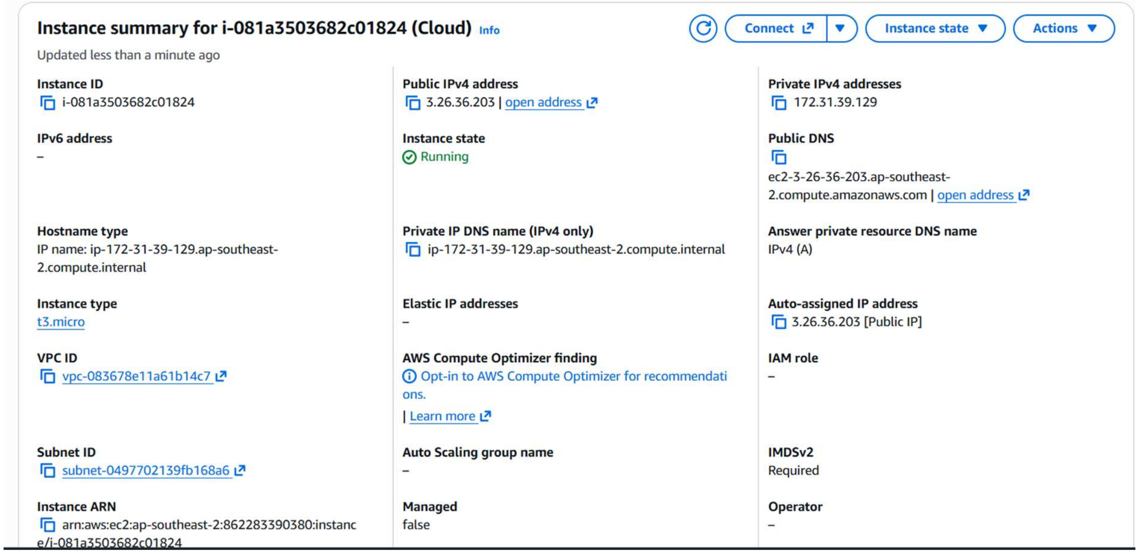
Cloud Challenges

1. Create a Ec2 instance from amazon linux 2



Instance summary for i-081a3503682c01824 (Cloud)

Updated less than a minute ago

Instance ID
i-081a3503682c01824

Public IPv4 address
3.26.36.203 | open address

Private IPv4 addresses
172.31.39.129

IPv6 address
–

Instance state
Running

Public DNS
ec2-3-26-36-203.ap-southeast-2.compute.amazonaws.com | open address

Hostname type
IP name: ip-172-31-39-129.ap-southeast-2.compute.internal

Private IP DNS name (IPv4 only)
ip-172-31-39-129.ap-southeast-2.compute.internal

Answer private resource DNS name
IPv4 (A)

Instance type
t3.micro

Elastic IP addresses
–

Auto-assigned IP address
3.26.36.203 [Public IP]

VPC ID
vpc-083678e11a61b14c7

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
| Learn more

IAM role
–

Subnet ID
subnet-0497702139fb168a6

Auto Scaling group name
–

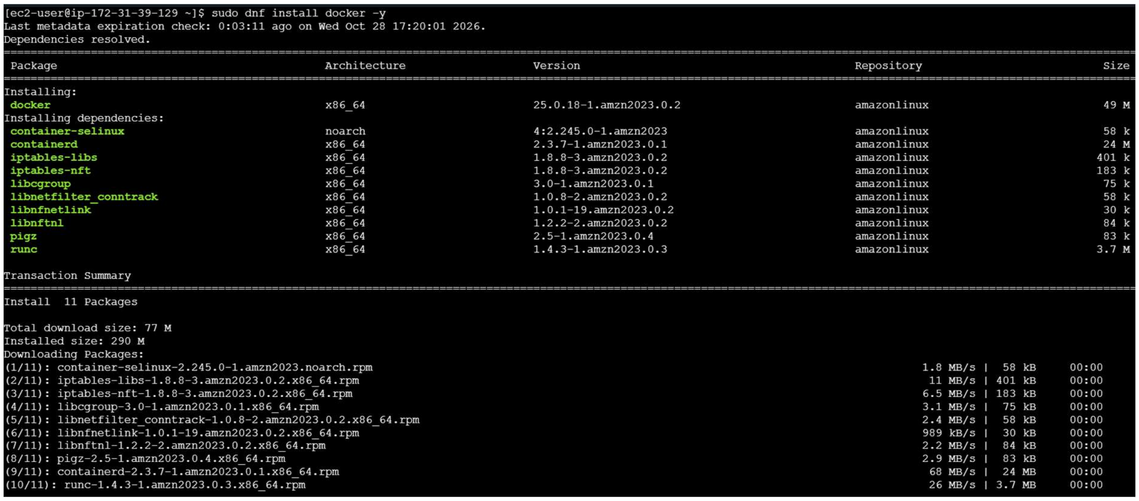
IMDSv2
Required

Instance ARN
arn:aws:ec2:ap-southeast-2:862283390380:instance/i-081a3503682c01824

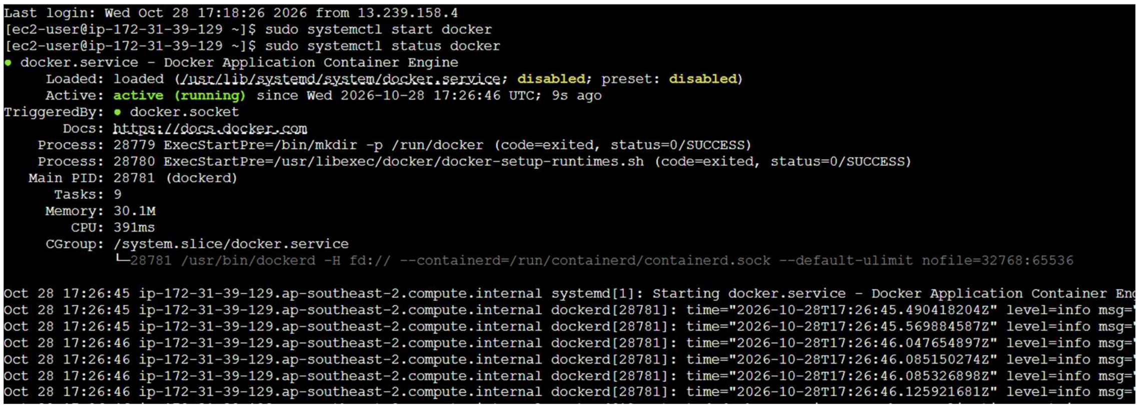
Managed
false

Operator
–

2. Install Docker



```
[ec2-user@ip-172-31-39-129 ~]$ sudo dnf install docker -y
Last metadata expiration check: 0:03:11 ago on Wed Oct 28 17:20:01 2026.
Dependencies resolved.
==========================================================================================================================================================
 Package                             Architecture              Version                                      Repository                               Size
==========================================================================================================================================================
Installing:
 docker                              x86_64                    25.0.18-1.amzn2023.0.2                       amazonlinux                              49 M
Installing dependencies:
 container-selinux                   noarch                    4:2.245.0-1.amzn2023                         amazonlinux                              58 k
 containerd                          x86_64                    2.3.7-1.amzn2023.0.1                         amazonlinux                              24 M
 iptables-libs                       x86_64                    1.8.8-3.amzn2023.0.2                         amazonlinux                             401 k
 iptables-nft                        x86_64                    1.8.8-3.amzn2023.0.2                         amazonlinux                             183 k
 libcgroup                           x86_64                    3.0-1.amzn2023.0.1                           amazonlinux                              75 k
 libnetfilter_conntrack              x86_64                    1.0.8-2.amzn2023.0.2                         amazonlinux                              58 k
 libnfnetlink                        x86_64                    1.0.1-19.amzn2023.0.2                        amazonlinux                              30 k
 libnftnl                            x86_64                    1.2.2-2.amzn2023.0.2                         amazonlinux                              84 k
 pigz                                x86_64                    2.5-1.amzn2023.0.4                           amazonlinux                              83 k
 runc                                x86_64                    1.4.3-1.amzn2023.0.3                         amazonlinux                             3.7 M

Transaction Summary
==========================================================================================================================================================
Install  11 Packages

Total download size: 77 M
Installed size: 290 M
Downloading Packages:
(1/11): container-selinux-2.245.0-1.amzn2023.noarch.rpm                                                                  1.8 MB/s |  58 kB     00:00
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm                                                                     11 MB/s | 401 kB     00:00
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm                                                                     6.5 MB/s | 183 kB     00:00
(4/11): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm                                                                          3.1 MB/s |  75 kB     00:00
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm                                                           2.4 MB/s |  58 kB     00:00
(6/11): libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64.rpm                                                                    989 kB/s |  30 kB     00:00
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm                                                                         2.2 MB/s |  84 kB     00:00
(8/11): pigz-2.5-1.amzn2023.0.4.x86_64.rpm                                                                               2.9 MB/s |  83 kB     00:00
(9/11): containerd-2.3.7-1.amzn2023.0.1.x86_64.rpm                                                                        68 MB/s |  24 MB     00:00
(10/11): runc-1.4.3-1.amzn2023.0.3.x86_64.rpm                                                                             26 MB/s | 3.7 MB     00:00
```



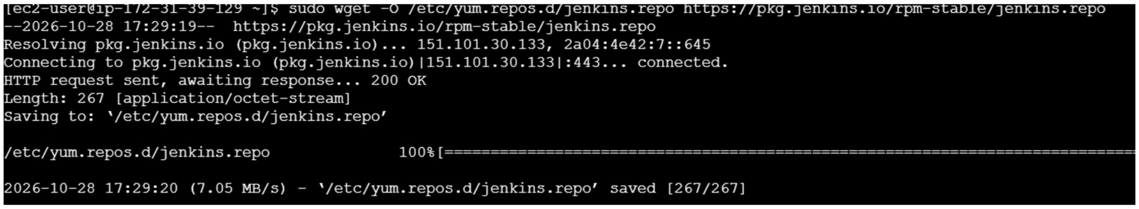
```
Last login: Wed Oct 28 17:18:26 2026 from 13.239.158.4
[ec2-user@ip-172-31-39-129 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-39-129 ~]$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-10-28 17:26:46 UTC; 9s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
    Process: 28779 ExecStartPre=/bin/mkdir -p /run/docker (code=exited, status=0/SUCCESS)
    Process: 28780 ExecStartPre=/usr/libexec/docker/docker-setup-runtimes.sh (code=exited, status=0/SUCCESS)
   Main PID: 28781 (dockerd)
      Tasks: 9
     Memory: 30.1M
        CPU: 391ms
     CGroup: /system.slice/docker.service
             └─28781 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536

Oct 28 17:26:45 ip-172-31-39-129.ap-southeast-2.compute.internal systemd[1]: Starting docker.service - Docker Application Container En
Oct 28 17:26:45 ip-172-31-39-129.ap-southeast-2.compute.internal dockerd[28781]: time="2026-10-28T17:26:45.490418204Z" level=info msg=
Oct 28 17:26:45 ip-172-31-39-129.ap-southeast-2.compute.internal dockerd[28781]: time="2026-10-28T17:26:45.569884587Z" level=info msg=
Oct 28 17:26:46 ip-172-31-39-129.ap-southeast-2.compute.internal dockerd[28781]: time="2026-10-28T17:26:46.047654897Z" level=info msg=
Oct 28 17:26:46 ip-172-31-39-129.ap-southeast-2.compute.internal dockerd[28781]: time="2026-10-28T17:26:46.085150274Z" level=info msg=
Oct 28 17:26:46 ip-172-31-39-129.ap-southeast-2.compute.internal dockerd[28781]: time="2026-10-28T17:26:46.085326898Z" level=info msg=
Oct 28 17:26:46 ip-172-31-39-129.ap-southeast-2.compute.internal dockerd[28781]: time="2026-10-28T17:26:46.125921681Z" level=info msg=
```

3.Install Jenkins

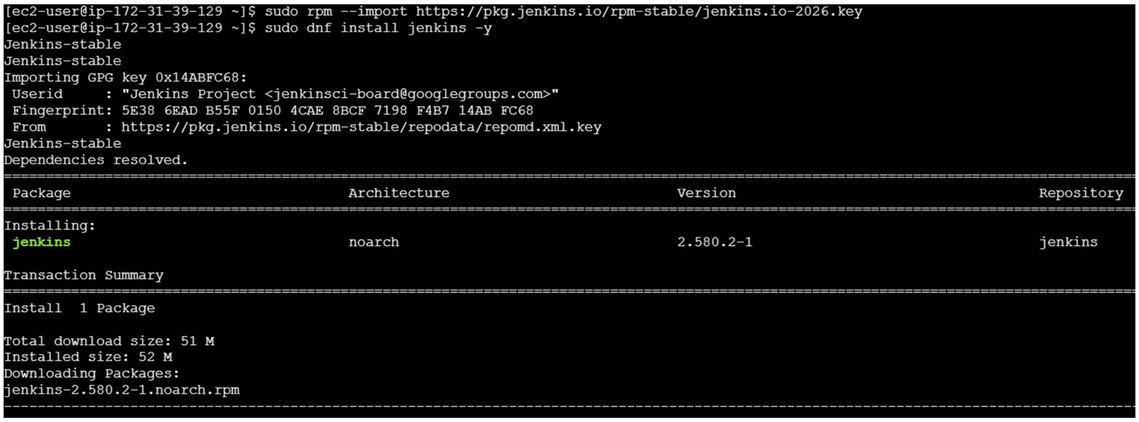
Step 1:- First install the Jenkins repository

Step 2:- Import the Jenkins key

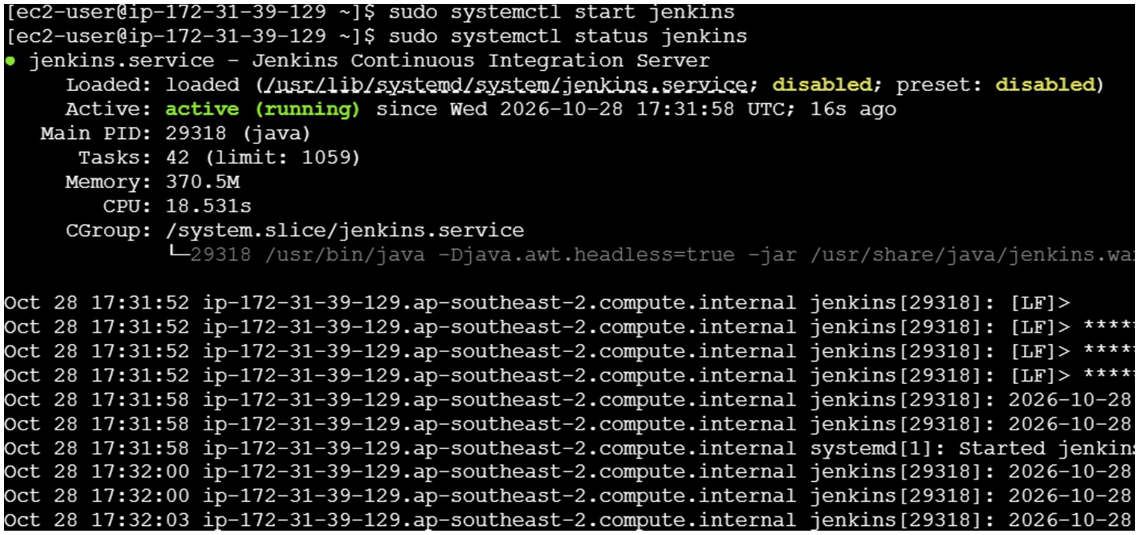
Step 3:- Then install the Jenkins



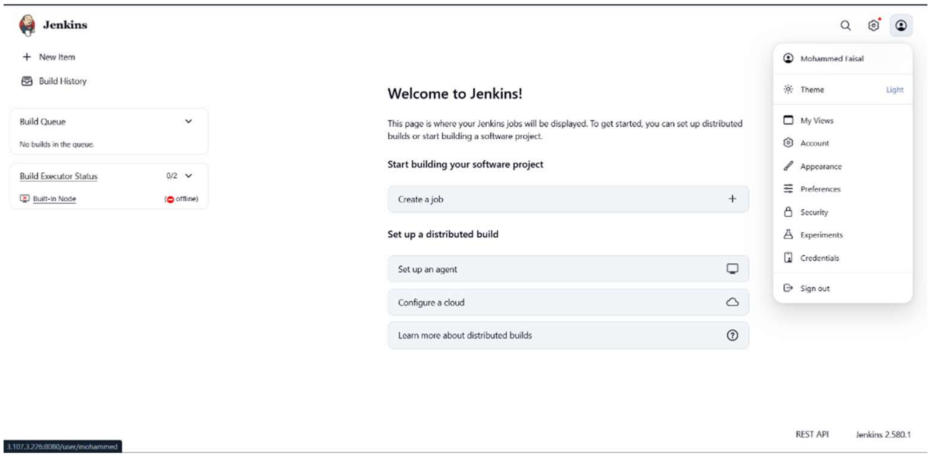
```
[ec2-user@ip-172-31-39-129 ~]$ sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/rpm-stable/jenkins.repo
--2026-10-28 17:29:19--  https://pkg.jenkins.io/rpm-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 151.101.30.133, 2a04:4e42:7::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|151.101.30.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 267 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenkins.repo             100%[=======================================================================

2026-10-28 17:29:20 (7.05 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [267/267]
```



```
[ec2-user@ip-172-31-39-129 ~]$ sudo rpm --import https://pkg.jenkins.io/rpm-stable/jenkins.io-2026.key
[ec2-user@ip-172-31-39-129 ~]$ sudo dnf install jenkins -y
Jenkins-stable
Jenkins-stable
Importing GPG key 0x14ABFC68:
 Userid     : "Jenkins Project <jenkinsci-board@googlegroups.com>"
 Fingerprint: 5E38 6EAD B55F 0150 4CAE 8BCF 7198 F4B7 14AB FC68
 From       : https://pkg.jenkins.io/rpm-stable/repodata/repomd.xml.key
Jenkins-stable
Dependencies resolved.
==================================================================================================================================
 Package                                 Architecture                           Version                                Repository
==================================================================================================================================
Installing:
 jenkins                                 noarch                                 2.580.2-1                              jenkins

Transaction Summary
==================================================================================================================================
Install  1 Package

Total download size: 51 M
Installed size: 52 M
Downloading Packages:
jenkins-2.580.2-1.noarch.rpm
----------------------------------------------------------------------------------------------------------------------------------
```



```
[ec2-user@ip-172-31-39-129 ~]$ sudo systemctl start jenkins
[ec2-user@ip-172-31-39-129 ~]$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-10-28 17:31:58 UTC; 16s ago
   Main PID: 29318 (java)
      Tasks: 42 (limit: 1059)
     Memory: 370.5M
        CPU: 18.531s
     CGroup: /system.slice/jenkins.service
             └─29318 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.wa

Oct 28 17:31:52 ip-172-31-39-129.ap-southeast-2.compute.internal jenkins[29318]: [LF]>
Oct 28 17:31:52 ip-172-31-39-129.ap-southeast-2.compute.internal jenkins[29318]: [LF]> ****
Oct 28 17:31:52 ip-172-31-39-129.ap-southeast-2.compute.internal jenkins[29318]: [LF]> ****
Oct 28 17:31:52 ip-172-31-39-129.ap-southeast-2.compute.internal jenkins[29318]: [LF]> ****
Oct 28 17:31:58 ip-172-31-39-129.ap-southeast-2.compute.internal jenkins[29318]: 2026-10-28
Oct 28 17:31:58 ip-172-31-39-129.ap-southeast-2.compute.internal jenkins[29318]: 2026-10-28
Oct 28 17:31:58 ip-172-31-39-129.ap-southeast-2.compute.internal systemd[1]: Started jenkin
Oct 28 17:32:00 ip-172-31-39-129.ap-southeast-2.compute.internal jenkins[29318]: 2026-10-28
Oct 28 17:32:00 ip-172-31-39-129.ap-southeast-2.compute.internal jenkins[29318]: 2026-10-28
Oct 28 17:32:03 ip-172-31-39-129.ap-southeast-2.compute.internal jenkins[29318]: 2026-10-28
```



Jenkins
New Item
Build History
Build Queue
Build Executor Status
0/2
Built-In Node
offline
Mohammed Faisal
Theme
Light
My Views
Account
Appearance
Preferences
Security
Experiments
Credentials
Sign out
Create a job
Set up an agent
Configure a cloud
Learn more about distributed builds
REST API
Jenkins 2.580.1
3.107.3.226:8080/user/mohammed

Welcome to Jenkins!

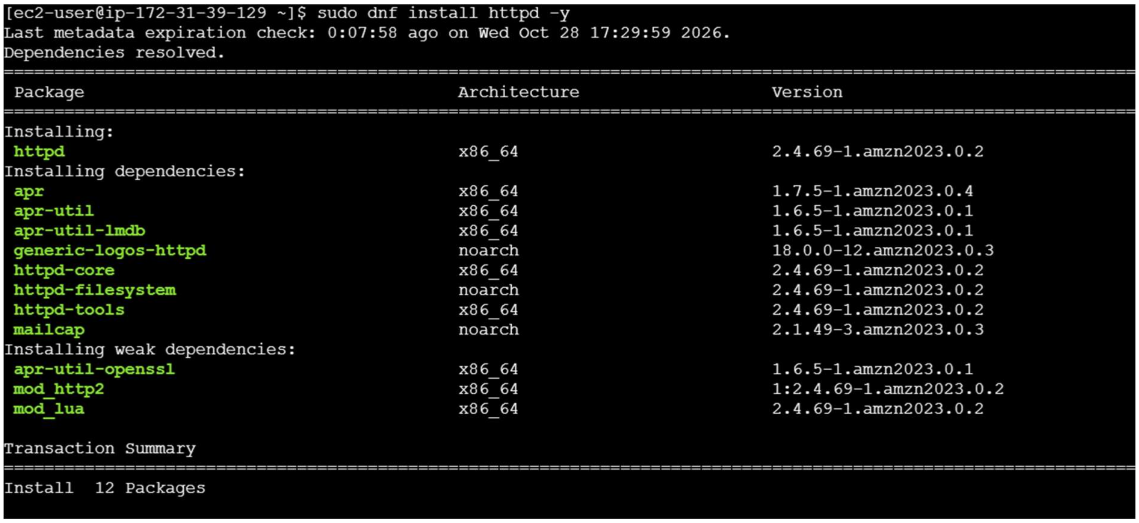
This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

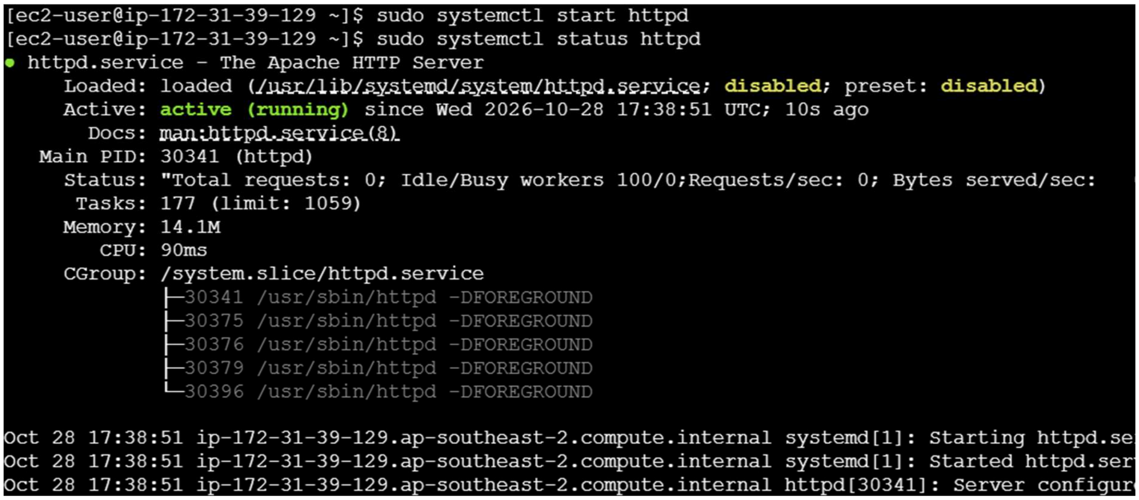
Set up a distributed build

No builds in the queue.

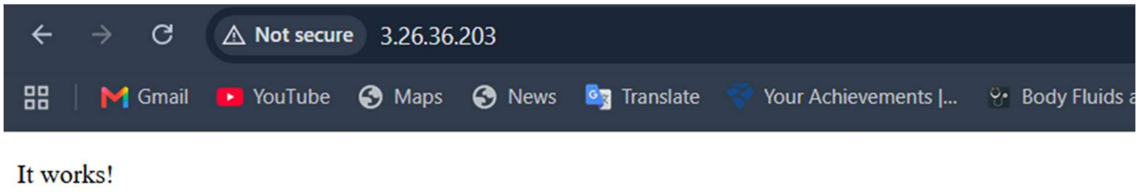
4. Install Apache



```
[ec2-user@ip-172-31-39-129 ~]$ sudo dnf install httpd -y
Last metadata expiration check: 0:07:58 ago on Wed Oct 28 17:29:59 2026.
Dependencies resolved.
================================================================================================================
 Package                                      Architecture                   Version
================================================================================================================
Installing:
 httpd                                        x86_64                         2.4.69-1.amzn2023.0.2
Installing dependencies:
 apr                                          x86_64                         1.7.5-1.amzn2023.0.4
 apr-util                                     x86_64                         1.6.5-1.amzn2023.0.1
 apr-util-lmdb                                x86_64                         1.6.5-1.amzn2023.0.1
 generic-logos-httpd                          noarch                         18.0.0-12.amzn2023.0.3
 httpd-core                                   x86_64                         2.4.69-1.amzn2023.0.2
 httpd-filesystem                             noarch                         2.4.69-1.amzn2023.0.2
 httpd-tools                                  x86_64                         2.4.69-1.amzn2023.0.2
 mailcap                                      noarch                         2.1.49-3.amzn2023.0.3
Installing weak dependencies:
 apr-util-openssl                             x86_64                         1.6.5-1.amzn2023.0.1
 mod_http2                                    x86_64                         1:2.4.69-1.amzn2023.0.2
 mod_lua                                      x86_64                         2.4.69-1.amzn2023.0.2

Transaction Summary
================================================================================================================
Install  12 Packages
```

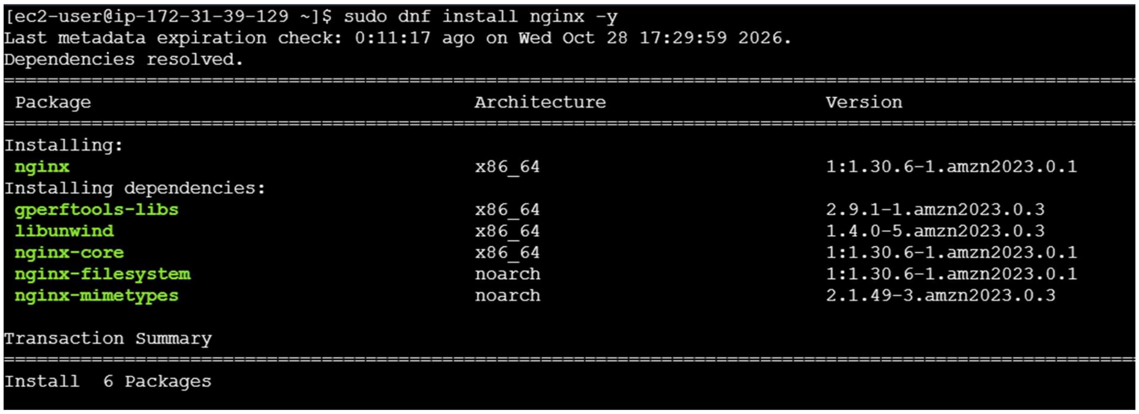


```
[ec2-user@ip-172-31-39-129 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-39-129 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-10-28 17:38:51 UTC; 10s ago
       Docs: man:httpd.service(8)
   Main PID: 30341 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:
      Tasks: 177 (limit: 1059)
     Memory: 14.1M
        CPU: 90ms
     CGroup: /system.slice/httpd.service
             ├─30341 /usr/sbin/httpd -DFOREGROUND
             ├─30375 /usr/sbin/httpd -DFOREGROUND
             ├─30376 /usr/sbin/httpd -DFOREGROUND
             ├─30379 /usr/sbin/httpd -DFOREGROUND
             └─30396 /usr/sbin/httpd -DFOREGROUND

Oct 28 17:38:51 ip-172-31-39-129.ap-southeast-2.compute.internal systemd[1]: Starting httpd.se
Oct 28 17:38:51 ip-172-31-39-129.ap-southeast-2.compute.internal systemd[1]: Started httpd.ser
Oct 28 17:38:51 ip-172-31-39-129.ap-southeast-2.compute.internal httpd[30341]: Server configur
```

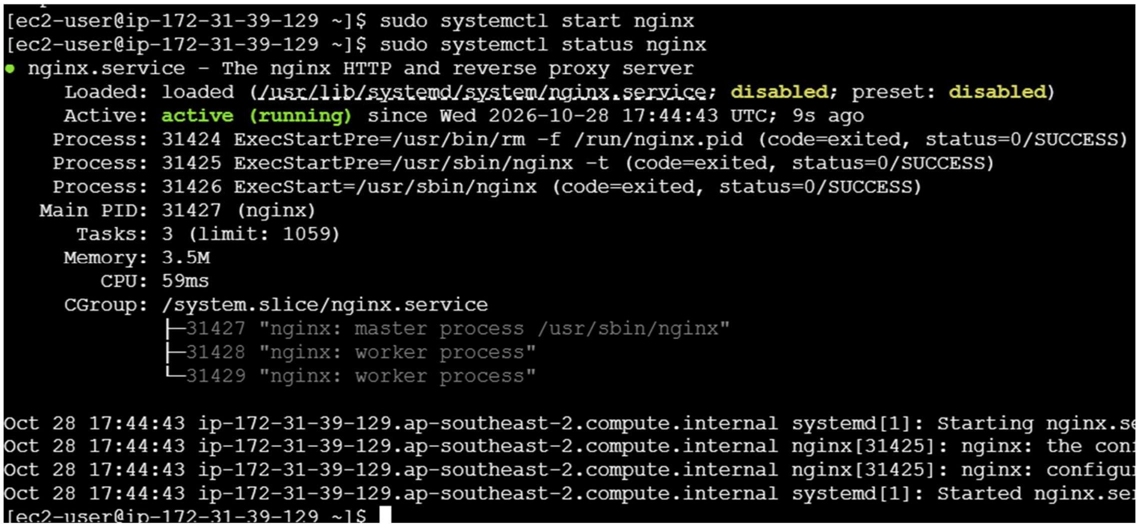


It works!

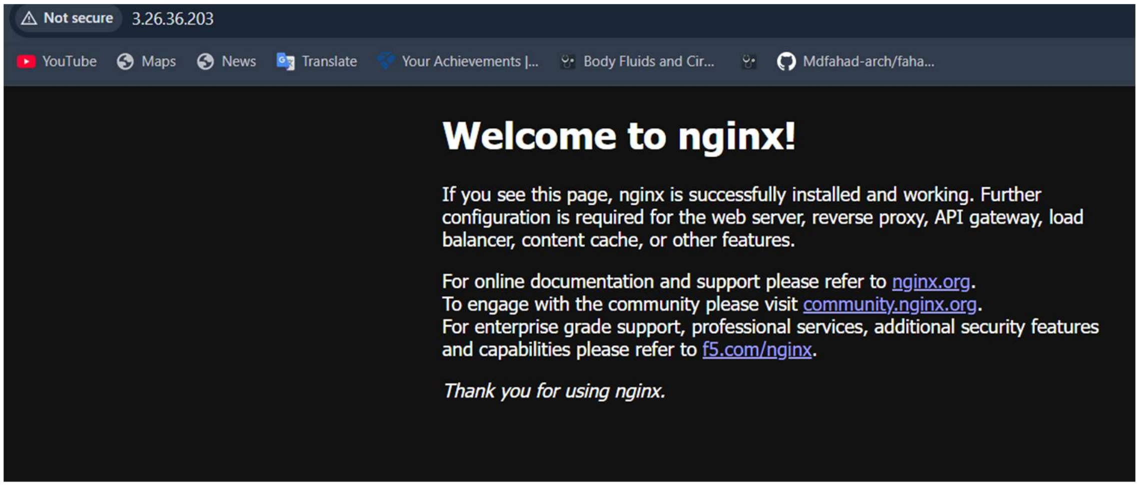
5.Install Nginx



```
[ec2-user@ip-172-31-39-129 ~]$ sudo dnf install nginx -y
Last metadata expiration check: 0:11:17 ago on Wed Oct 28 17:29:59 2026.
Dependencies resolved.
==========================================================================================================
 Package                                    Architecture                     Version
==========================================================================================================
Installing:
 nginx                                      x86_64                           1:1.30.6-1.amzn2023.0.1
Installing dependencies:
 gperftools-libs                            x86_64                           2.9.1-1.amzn2023.0.3
 libunwind                                  x86_64                           1.4.0-5.amzn2023.0.3
 nginx-core                                 x86_64                           1:1.30.6-1.amzn2023.0.1
 nginx-filesystem                           noarch                           1:1.30.6-1.amzn2023.0.1
 nginx-mimetypes                            noarch                           2.1.49-3.amzn2023.0.3

Transaction Summary
==========================================================================================================
Install  6 Packages
```



```
[ec2-user@ip-172-31-39-129 ~]$ sudo systemctl start nginx
[ec2-user@ip-172-31-39-129 ~]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-10-28 17:44:43 UTC; 9s ago
    Process: 31424 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Process: 31425 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
    Process: 31426 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 31427 (nginx)
      Tasks: 3 (limit: 1059)
     Memory: 3.5M
        CPU: 59ms
     CGroup: /system.slice/nginx.service
             ├─31427 "nginx: master process /usr/sbin/nginx"
             ├─31428 "nginx: worker process"
             └─31429 "nginx: worker process"

Oct 28 17:44:43 ip-172-31-39-129.ap-southeast-2.compute.internal systemd[1]: Starting nginx.se
Oct 28 17:44:43 ip-172-31-39-129.ap-southeast-2.compute.internal nginx[31425]: nginx: the con
Oct 28 17:44:43 ip-172-31-39-129.ap-southeast-2.compute.internal nginx[31425]: nginx: configu
Oct 28 17:44:43 ip-172-31-39-129.ap-southeast-2.compute.internal systemd[1]: Started nginx.se
[ec2-user@ip-172-31-39-129 ~]$
```



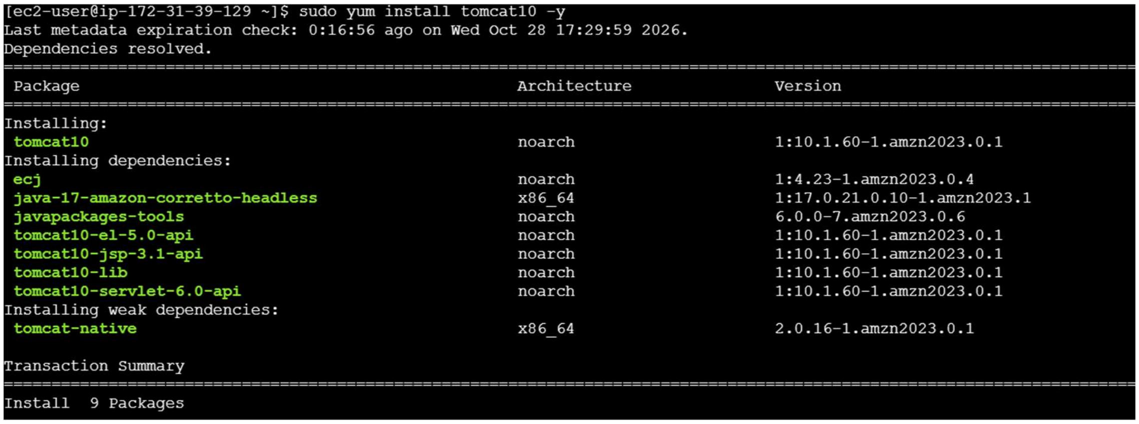
Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

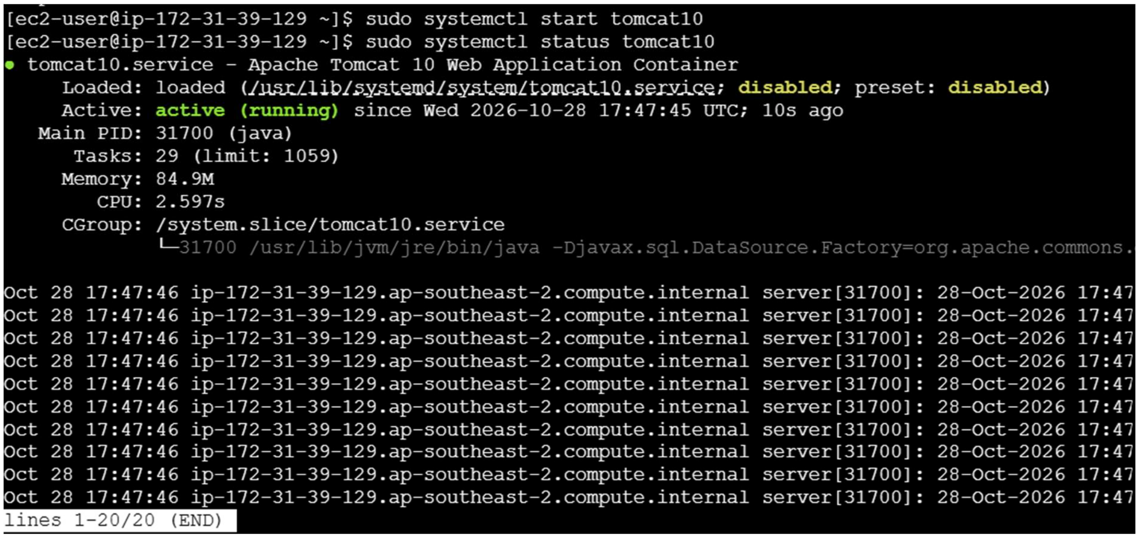
For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

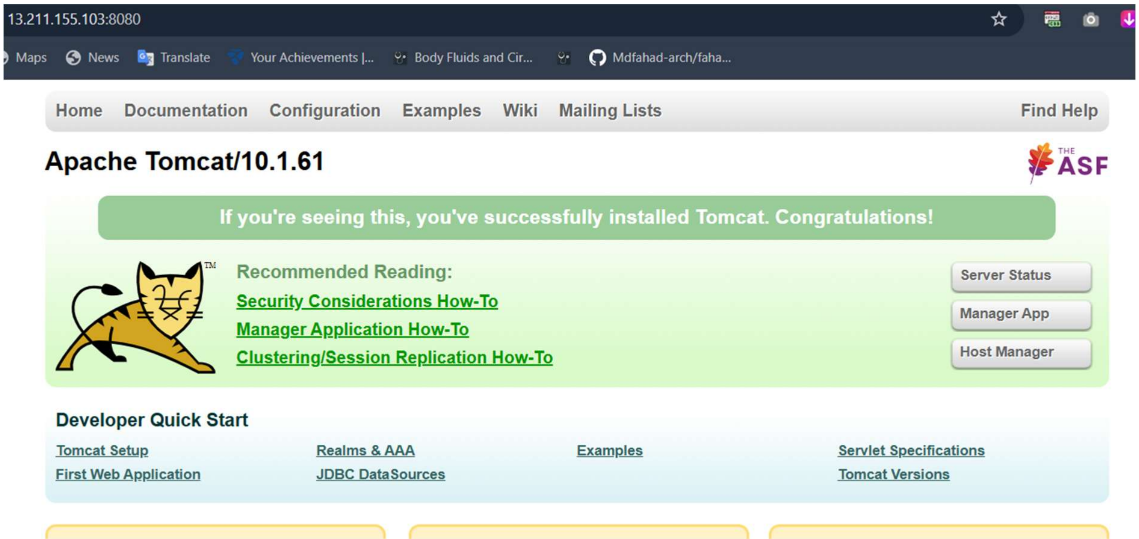
6.Install Apache Tomcat



```
[ec2-user@ip-172-31-39-129 ~]$ sudo yum install tomcat10 -y
Last metadata expiration check: 0:16:56 ago on Wed Oct 28 17:29:59 2026.
Dependencies resolved.
========================================================================================================================
 Package                                                Architecture              Version
========================================================================================================================
Installing:
 tomcat10                                               noarch                    1:10.1.60-1.amzn2023.0.1
Installing dependencies:
 ecj                                                    noarch                    1:4.23-1.amzn2023.0.4
 java-17-amazon-corretto-headless                       x86_64                    1:17.0.21.0.10-1.amzn2023.1
 javapackages-tools                                     noarch                    6.0.0-7.amzn2023.0.6
 tomcat10-el-5.0-api                                    noarch                    1:10.1.60-1.amzn2023.0.1
 tomcat10-jsp-3.1-api                                   noarch                    1:10.1.60-1.amzn2023.0.1
 tomcat10-lib                                           noarch                    1:10.1.60-1.amzn2023.0.1
 tomcat10-servlet-6.0-api                               noarch                    1:10.1.60-1.amzn2023.0.1
Installing weak dependencies:
 tomcat-native                                          x86_64                    2.0.16-1.amzn2023.0.1

Transaction Summary
========================================================================================================================
Install  9 Packages
```



```
[ec2-user@ip-172-31-39-129 ~]$ sudo systemctl start tomcat10
[ec2-user@ip-172-31-39-129 ~]$ sudo systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Container
     Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-10-28 17:47:45 UTC; 10s ago
   Main PID: 31700 (java)
      Tasks: 29 (limit: 1059)
     Memory: 84.9M
        CPU: 2.597s
     CGroup: /system.slice/tomcat10.service
             └─31700 /usr/lib/jvm/jre/bin/java -Djavax.sql.DataSource.Factory=org.apache.commons.

Oct 28 17:47:46 ip-172-31-39-129.ap-southeast-2.compute.internal server[31700]: 28-Oct-2026 17:47
Oct 28 17:47:46 ip-172-31-39-129.ap-southeast-2.compute.internal server[31700]: 28-Oct-2026 17:47
Oct 28 17:47:46 ip-172-31-39-129.ap-southeast-2.compute.internal server[31700]: 28-Oct-2026 17:47
Oct 28 17:47:46 ip-172-31-39-129.ap-southeast-2.compute.internal server[31700]: 28-Oct-2026 17:47
Oct 28 17:47:46 ip-172-31-39-129.ap-southeast-2.compute.internal server[31700]: 28-Oct-2026 17:47
Oct 28 17:47:46 ip-172-31-39-129.ap-southeast-2.compute.internal server[31700]: 28-Oct-2026 17:47
Oct 28 17:47:46 ip-172-31-39-129.ap-southeast-2.compute.internal server[31700]: 28-Oct-2026 17:47
Oct 28 17:47:46 ip-172-31-39-129.ap-southeast-2.compute.internal server[31700]: 28-Oct-2026 17:47
Oct 28 17:47:46 ip-172-31-39-129.ap-southeast-2.compute.internal server[31700]: 28-Oct-2026 17:47
Oct 28 17:47:46 ip-172-31-39-129.ap-southeast-2.compute.internal server[31700]: 28-Oct-2026 17:47
lines 1-20/20 (END)
```



Apache Tomcat/10.1.61

If you're seeing this, you've successfully installed Tomcat. Congratulations!

Recommended Reading:

Security Considerations How-To

Manager Application How-To

Clustering/Session Replication How-To

Developer Quick Start

Tomcat Setup

First Web Application

Realms & AAA

JDBC DataSources

Examples

Servlet Specifications

Tomcat Versions